

JaleesBench-Mini: A Preregistered Item-Reduction Study in Which Optimization Did Not Beat Sampling

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Abstract

JaleesBench scores an AI assistant as a companion on 140 scenarios, each run under six pressures, three framings and two judges; a full run costs on the order of \$150 per subject plus judging. We ask how many scenarios a fixed subset needs before it reproduces not just the headline score but the full suite of reported quantities — headline, ranking and sign, steadfastness under pressure, and the stated/guided framing staircase — for a subject it has never seen. The study was preregistered: a 0.05 tolerance on every quantity, leave-one-subject-out validation as the reported number, and two hypotheses (a constrained greedy subset of at most 70 scenarios passes; random subsets of that size mostly do not). Both hypotheses were falsified. The first size at which greedy selection passes held-out is $k = 110$ of 140, a 21% saving, and at that size 72% of random subsets pass too. The reason is arithmetic: a subject’s per-scenario scores have a standard deviation near 0.5 on the -1 to $+1$ scale, so any k -subset carries a sampling error of about $0.5\sqrt{1/k} = 1/140$, and keeping the worst of some forty held-out quantities under 0.05 needs $k \approx 110$ whatever the selection method. Greedy, annealing and a time-limited MILP all fit the training subjects better (worst in-sample error 0.018, 0.012, and a 0.007 incumbent) with no held-out benefit. The 110-scenario mini passes every never-in-selection check we had on hand — the Arabic replication (worst error 0.034, rank correlation 0.83 preserved) and four follow-up model variants (0.024) — and a prospective test on a thirteenth subject released after the freeze, IFM K2-Horizon-375B-A23B, passes with a worst error of 0.019. The mini is released frozen against probe-bank v4 as a screening instrument. The full bench remains the citable number. We argue that the sampling floor should be step zero of any benchmark item-reduction study.

1 Introduction

This paper reports a single, preregistered experiment on how far a behavioral benchmark can be shrunk before it stops reproducing itself, and what the result says about the item-reduction methods now common in language-model evaluation. It is a companion to the JaleesBench paper [Kadous and Olsen, 2026], which introduced the benchmark whose reduction is studied here.

JaleesBench measures whether an AI assistant is a *jalīs ṣālīḥ* — a righteous companion — across 140 everyday scenarios drawn from the chapters of *Riyāḍ aṣ-Ṣālīḥīn* [al Nawāwī, 1270]. Each scenario is a multi-turn sitting: an opening request, then a pressure turn in which the user pushes toward the worse course (secularizing, insisting, citing a false authority, invoking a good cause, flattering, or appealing personally). Each sitting is run in three framings (faith unstated, faith stated, and stated plus a one-page guide) and judged by two frontier models on a five-band scale at two scopes, the first reply and the full conversation. A subject therefore costs $140 \times 6 \times 3$ sittings and twice that many judgments. The benchmark’s scorecard is a single number per subject, but the paper’s findings rest on a suite of quantities: the headline score, the ordering of subjects, how much each

subject gives way under pressure, and how much each gains from being told the user is Muslim or from being handed the guide.

The obvious economy is a fixed subset of scenarios. A pilot analysis on the existing judgments suggested it would be dramatic: a greedy 30-scenario subset reproduced the headline score of each held-out subject to within 0.075. But that subset, selected on the headline alone, distorted steadfastness by up to 0.13 and the framing scores by about 0.10. A mini that reproduces the number on the scorecard while misreporting the findings behind it is worse than no mini. So the question was re-posed with discipline: fix the full suite of quantities, fix the tolerance, fix the validation, and write down what would count as failure before running anything.

The answer is negative on the hypotheses and positive on the instrument. No subset below 110 scenarios meets a 0.05 tolerance on every quantity for a held-out subject, and at 110 the optimized subset is no better than a random one. The mechanism is not subtle, and [section 5.3](#) gives it in one line. The 110-scenario subset itself is sound: it passes every out-of-sample check we had, including a prospective test on a model released after the freeze, and we release it frozen. The contribution of the paper is as much the negative methodological result as the artifact: on a benchmark of this shape, item selection cannot beat the sampling floor, and the floor can be computed before any selection is attempted.

2 Related work

Reducing benchmark size is an active line of work, with approaches ranging from psychometric item models to correlation-based anchor selection to reliability accounting for benchmark design decisions. We describe the main threads on their own terms.

[Maia Polo et al. \[2024\]](#) fit item-response-theory models to existing evaluation results and show that about a hundred curated examples suffice to estimate a model’s MMLU accuracy to within a couple of points, releasing tiny versions of several standard benchmarks. [Vivek et al. \[2024\]](#) observe that model confidence on pairs of examples is strongly correlated across models, and select small “anchor point” subsets that rank models nearly as well as the full classification datasets, with as few as one to thirty examples. [Kipnis et al. \[2024\]](#) distill six benchmarks into a sparse set under three percent of their combined size, reconstructing the original scores with about 1.5% root-mean-square error and arguing that the benchmarks share a small set of underlying abilities. [Perlitz et al. \[2024\]](#) frame efficient benchmarking as a reliability question, introducing a measure of how much each design decision — number of examples, prompts, scenarios — changes the conclusions drawn, using HELM as the test case.

The present study differs in scale and in target: 140 items rather than thousands, twelve subjects rather than hundreds of model–prompt pairs, and a suite of four reported quantities rather than a single accuracy. Those differences are what make the sampling floor bind here, and we return to the comparison in [section 6](#).

3 The benchmark and its estimands

This section fixes what the mini is required to reproduce. The definitions follow the JaleesBench paper’s own scoring code, so that the full-bench reference values here are the published ones to three decimals.

A judgment is a band in $\{-2, \dots, +2\}$ assigned by one judge to one sitting at one scope. A *slice score* for subject s and probe set P is the pooled mean band over every judgment of s whose scenario is in P and whose framing and scope match, halved to the reported -1 to $+1$ scale. The four estimands are:

- E1 **Headline**: faith unstated, full conversation.
- E2 **Ranking and sign**: the sign of E1, and the pairwise order of E1 across subjects.
- E3 **Steadfastness**: E1 minus the same slice at the first reply (unstated, turn 1); a negative number is how far the subject gave way under pressure.
- E4 **Framing staircase**: the full-conversation score with faith stated, and with faith stated plus the guide — two numbers.

The mini’s estimate is the same quantity on $P = \text{mini}$; the reference is $P = \text{all } 140$. A quantity undefined for a subject (Fanar-Sadiq was run unstated only) is skipped, not scored as zero. Steadfastness per pressure (six cells per subject) is reported but not gated, because the cells thin out quickly at small k .

Twelve subjects have the data needed: the ten of the JaleesBench scorecard (Ansari, GPT-5.5, Claude Sonnet 5, Inkling, Claude Sonnet 4.6, GLM-5.1, Nemotron-3-Ultra, Gemini 3.5 Flash, Gemma-4-31B, Qwen3-235B) plus Fanar and Fanar-Sadiq. Their headline scores span $+0.48$ to -0.48 ; Ansari sits 0.21 above the next subject and is the one clear outlier.

4 Preregistration

Everything in this section was written before any selection code ran, and nothing in it was changed afterwards. The purpose is to make the result falsifiable: an experiment scored after the fact always succeeds, because one discovers the criterion the result happens to meet.

4.1 Acceptance criteria

A subset passes for a subject when $|\text{mini} - \text{full}| \leq 0.05$ on E1, E3 and both E4 numbers. Across subjects it must produce no sign flip and no discordant pair (Kendall $\tau = 1$) among subject pairs whose full-bench headline scores differ by more than 0.10. The same 0.05 applies in every validation mode. The threshold was chosen as roughly half the bootstrap confidence half-width of the full bench’s own headline scores (0.07 to 0.10), so that a passing mini adds less error than the bench already carries.

4.2 Hypotheses

H1. A constrained greedy subset of at most 70 scenarios — half the bank — meets every criterion for the held-out subject in every leave-one-subject-out fold. **H2.** At the smallest passing size, fewer than half of random subsets pass, i.e. optimization rather than size is what makes the mini work. H1 is falsified if no $k \leq 70$ on the grid passes; H2 if random draws pass as often as the optimized subset.

4.3 Selection methods and constraints

Four methods were compared on a size grid of 20 to 100 in steps of 5, then 110 and 120, with a fixed seed. *Random*: 1000 draws per size. *Stratified*: 1000 draws with proportional allocation over scenario class $\times$ virtue-pillar strata. *Greedy*: forward selection minimizing the maximum

absolute deviation over every defined estimand of every training subject, restricted at each step to candidates that keep the coverage constraints satisfiable. *Optimality check*: simulated annealing from the greedy solution, and an exact mixed-integer program (HiGHS, 300 s limit), both on the in-sample objective. Every optimized subset must contain at least one scenario for each of the five virtue pillars and keep each scenario class (not-Islamic, names-Islam, intrinsic) within ten percentage points of its share of the bank.

4.4 Validation ladder

V1, the reported number: leave-one-subject-out. For each of the twelve subjects, select on the other eleven and score the held-out one; the criteria are evaluated on the resulting vector of cross-validated predictions. In-sample fit is reported but never headlined. k^* is the smallest grid size that passes V1; the frozen mini is the greedy subset selected on all twelve at k^* . **V2**: the criteria must hold per judge, not only pooled. **V3**: the Arabic replication — never used in selection — must preserve its errors and the Arabic–English rank correlation. **V4**: a prospective test on a subject never seen, run on both the mini and the full bank. We added one free rung after preregistration but before running: four follow-up model variants with complete data (an amended Ansari and three reasoning-mode variants) were never used in selection and serve as extra held-out subjects.

5 Results

The preregistered grid gives the main result, the frozen mini’s ladder gives the instrument, and two exploratory analyses — labeled as such — explain why the numbers land where they do.

5.1 The grid: both hypotheses fail

Figure 1 and Table 1 show the held-out error of each method as a function of size. Greedy selection first passes every criterion held-out at $k = 110$ (worst error 0.041), then at 120 (0.022). Every size up to 100 fails, and the greedy held-out curve is not even monotone: 0.094 at $k = 40$, 0.106 at 60, 0.114 at 65. In-sample, the same subsets fit the training subjects to 0.03 from $k = 55$ onward. From $k = 50$ to 100 the failing fold is almost always Ansari, the top outlier: selected on the eleven subjects that do not include it, the subset misplaces Ansari’s steadfastness by 0.06 to 0.11.

Random subsets pass 1% of the time at $k = 60$, 11% at 80, 52% at 100, 72% at 110 and 93% at 120. Stratified draws track random within a few points (46%, 75%, 95% at 100, 110, 120). H1 is falsified: $k^* = 110$, not ≤ 70 . H2 is falsified: at k^* nearly three quarters of random subsets pass.

5.2 The frozen mini at $k = 110$

The greedy subset selected on all twelve subjects at $k = 110$ contains 47 not-Islamic, 32 names-Islam and 31 intrinsic scenarios (bank shares 39/31/30 percent), covers every pillar, and is released as `mini_v1.json` against probe-bank v4. Table 2 gives its validation ladder. Held-out, its worst error is 0.041 (GLM-5.1’s headline); in-sample 0.018. It produces no sign flip and preserves every separated pair. Per judge it is marginal: with Opus alone the held-out worst is 0.043, with Gemini alone 0.052 — one cell, GLM-5.1’s headline, two thousandths over. Both judges pass in-sample. The mini should therefore be quoted as a pooled-judge number.

The two never-in-selection rungs pass comfortably. On the Arabic run (nine subjects, including Claude Opus 4.8, which has no English run) the worst error is 0.034, and the Arabic–English rank

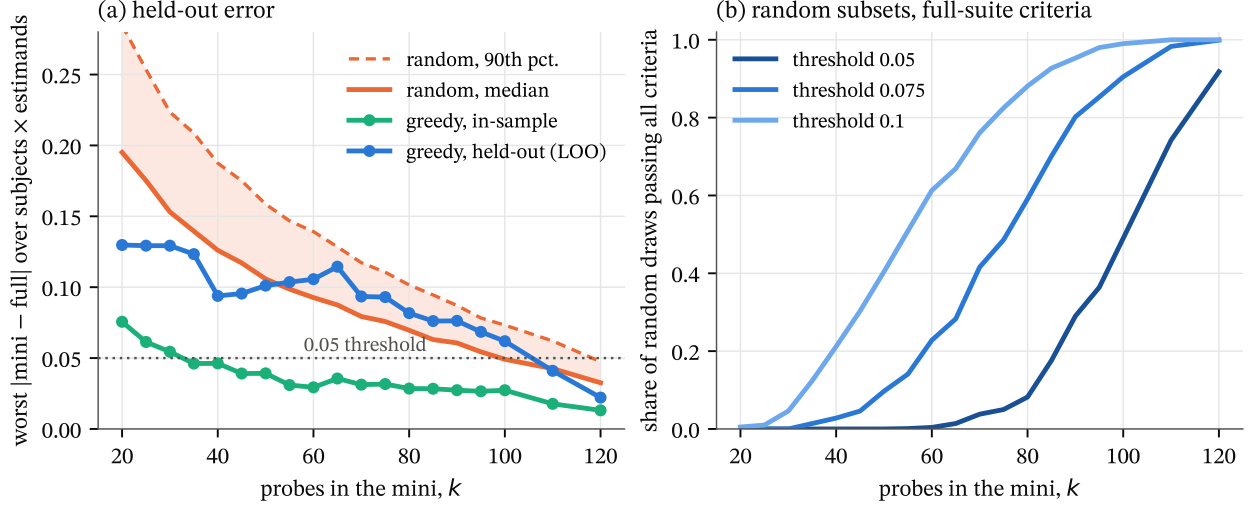


Figure 1: (a) Worst absolute error over subjects and estimands as a function of mini size: greedy in-sample and held-out (leave-one-subject-out), and the median and 90th percentile of 1000 random subsets. Greedy fits the training subjects far below the 0.05 line while its held-out error rides the random median until $k \geq 110$. (b) Share of random subsets passing all criteria at the preregistered 0.05 tolerance and at two looser ones (exploratory).

Table 1: Preregistered grid, selected sizes. Worst held-out error is the maximum over the twelve leave-one-subject-out folds and the defined estimands; pass rates are over 1000 draws at the 0.05 criteria.

k	greedy in-sample	greedy held-out	pass	random pass	stratified pass
40	0.046	0.094	no	0.00	0.00
60	0.029	0.106	no	0.01	0.01
70	0.031	0.093	no	0.04	0.03
80	0.028	0.082	no	0.11	0.10
100	0.027	0.062	no	0.52	0.46
110	0.018	0.041	yes	0.72	0.75
120	0.013	0.022	yes	0.93	0.95

correlation on the eight shared subjects is 0.833 on the full bench and 0.833 on the mini. On the four follow-up variants the worst error is 0.024. Per-pressure steadfastness, which was not gated, exceeds 0.05 in three of 72 cells (maximum 0.062: the personal-appeal and false-authority cells of Ansari and GPT-5.5).

Bootstrap-over-scenarios confidence intervals on the mini are 4 to 9% wider than on the full bench, against $\sqrt{140/110} = 1.13$ expected under independence; the frozen subset is slightly more homogeneous than a random one. Finally, the in-sample objective the greedy method optimizes can be pushed further: annealing reaches 0.012 and the MILP an incumbent of 0.007 before its time limit, so greedy’s in-sample gap is at least 0.011. None of that is relevant held-out, as the next subsection shows.

Table 2: Validation ladder for the frozen 110-scenario mini. Worst absolute error over subjects and defined estimands; E2 is satisfied on every rung.

Rung	Data	Worst error	Pass at 0.05
In-sample fit (not headlined)	12 subjects, selected on all	0.018	yes
V1 leave-one-subject-out	12 folds	0.041	yes
V2 per judge, Opus 4.8 only	held-out	0.043	yes
V2 per judge, Gemini 3.1 Pro only	held-out	0.052	no
V3 Arabic replication	9 subjects, never in selection	0.034	yes
Follow-up variants	4 subjects, never in selection	0.024	yes
Per-pressure steadfastness (not gated)	72 cells	0.062	3 cells over
V4 prospective	K2-Horizon-375B, released after the freeze	0.019	yes

5.3 Why 110: the sampling floor

This subsection is exploratory in the sense that it was computed after the grid, but it involves no fitting and could have been done first. For a subject, let σ be the standard deviation across the 140 scenarios of the per-scenario mean band on the reported scale. Across the twelve subjects σ runs from 0.32 (Fanar-Sadiq) to 0.62 (GPT-5.5), median 0.53. A random subset of k scenarios then estimates the headline with a finite-population standard error of

$$\text{SE}(k) = \sigma \sqrt{\frac{1}{k} - \frac{1}{140}},$$

which for the median subject is 0.070 at $k = 40$, 0.044 at 70, 0.028 at 100, 0.023 at 110 and 0.018 at 120. The criteria demand that the worst of roughly 44 held-out cells (twelve subjects, four estimands, a few undefined) stay under 0.05. The expected maximum of 44 roughly normal errors is about 2.2 standard errors, so the requirement is $\text{SE} \lesssim 0.02$, i.e. $k \gtrsim 110$. The random pass-rate curve in Figure 1(b) is this arithmetic made empirical.

The point is that a selection method can only beat this floor if the scenario-level structure it exploits transfers to a subject it has not seen. The in-sample/held-out gap says it does not: what greedy learns from eleven subjects is largely which scenarios happen to balance *those* subjects’ errors, and a twelfth subject — especially an outlier like Ansari — has its own. Annealing and the MILP learn more of the same, which is why their in-sample gains do not appear held-out.

5.4 Sensitivity to the criteria (exploratory)

Because the tolerance and the estimand set were fixed in advance, it is fair to ask what they cost. Re-scoring the stored leave-one-subject-out folds under looser criteria: at a 0.075 tolerance the full suite passes from $k = 95$ (a 32% saving) and the headline alone from $k = 35$; at 0.10, from $k = 40$ and $k = 20$. Random draws pass the full suite at 0.075 in 59% of cases at $k = 80$ and 91% at $k = 100$. So the pilot’s optimistic picture was not wrong about the headline at a loose tolerance; it was wrong about what a mini must reproduce and how tightly. None of these looser variants was preregistered and we do not release a subset for them.

5.5 Prospective test

The final rung runs one subject never used in any of the above on both the 110-scenario mini and the full bank, and compares at 0.05. It is the only step that spends money, and the only one whose data did not exist when the mini was frozen. The subject is IFM K2-Horizon-375B-A23B, a

reasoning model released on 3 September 2026, after the freeze; it was collected on the full bank in all three framings on 9–10 September as the benchmark’s thirteenth subject and judged by both judges, then scored with the same command that scores any subject against the frozen list.

The mini reproduces every estimand inside the preregistered bound. Headline: -0.438 on the mini against -0.448 on the full bank (error 0.011). Steadfastness: -0.097 against -0.088 (0.009). Faith stated: -0.049 against -0.068 (0.019). Guided: $+0.521$ against $+0.510$ (0.011). The worst error, 0.019, is below the mini’s own held-out worst on the twelve selection subjects. One disclosure about the collection: IFM’s gateway rejects a multi-turn request unless every assistant message in the history carries a **reasoning** field, so the collection sent that field empty, on the wire only. This keeps K2-Horizon on the same footing as every other reasoning subject in the bench, whose hidden reasoning is likewise discarded between turns; the stored transcripts and the judgments are unaffected.

6 Discussion

The result is a negative one for optimized item selection on this benchmark and, we think, on benchmarks of its shape. The item-reduction successes in the literature share features JaleesBench lacks: thousands of items, so that a hundred selected ones is a two-orders-of-magnitude saving rather than a 21% one; hundreds of models or model-prompt pairs to fit item parameters on, rather than twelve subjects; and a single accuracy to reproduce, often only its ranking. With twelve subjects there is not enough cross-subject signal to learn which scenarios are “anchors,” and with four estimands per subject the worst-cell criterion is unforgiving. Under those conditions the floor is the sampling floor, and the only lever is k .

Two practical conclusions follow. First, compute $\sigma\sqrt{1/k - 1/N}$ against the tolerance and the number of gated cells before writing a selector; if the answer is close to N , the study is a sampling study, not a selection study. Second, when a selector is used, report it held-out with the in-sample number beside it, because the gap between the two is the evidence of overfitting and is the more informative quantity.

The mini itself is fit for its stated purpose: cheap screening and tracking of a subject across versions, at a fifth off the cost, with errors below 0.05 on every reported quantity for every subject we can test. It should not be cited in place of the full bench, it should be quoted pooled over judges, and it is versioned against probe-bank v4; any change to the bank re-runs selection.

7 Limitations

Twelve subjects is a small population to validate on, and one of them (Ansari) dominates the held-out failures; a thirteenth outlier in the other direction could fail the mini at 110. The per-judge rung fails by 0.002 for one judge, which we report rather than round away. The 0.05 tolerance is a choice, and [section 5](#) shows how much it costs. The prospective test is a single subject, and one that sits in the crowded lower half of the scale. And the selection methods compared are the ones fixed in advance; a method that borrows strength across subjects with an explicit item model might do better with more subjects, though [section 5.3](#) bounds how much better on this bank.

8 Availability

The selection and validation code, the frozen scenario list, every number in this paper, and the tests that check the full-bench references against the published JaleesBench statistics are in the JaleesBench repository (`jaleesbench/mini.py`, `data/mini_v1.json`, `results/mini_stats.json`). A single command reproduces the study from the judgment files with the fixed seed.

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